

REPIF1 L6 Backup Server

User Story Documentation
SOFJO685 | School Year 2025/2026

1. Overview

This document covers the installation and configuration of the L6 Backup Server for the Portable Indoor Feedback (PIF) project. The backup server runs as a virtual machine and is responsible for creating daily backups of the project database hosted on the main web/database server, so that data can be recovered in the event of loss or a server crash.

2. Architecture

The backup infrastructure consists of:

- Web & Database Server Ubuntu 24.04 LTS, hosts Apache, MariaDB, PHP, and the project database (sofjo685_pif).
- Backup Server (this VM) Ubuntu 24.04 LTS, connects to the database server via MySQL client over the network and stores daily mysqldump exports.
- FTP Access allows external clients to retrieve backup files from the backup directory.

3. Requirements Status

The table below summarises all Must / Should / Could requirements from the User Story and their completion status.

Requirement	Priority	Status	Notes
Backup server installed & configured per work order	Must	✓ Done	Ubuntu 24.04 LTS VM
Daily DB backup at 08:00, stored on backup server	Must	✓ Done	cron + shell script
FTP access to backup directory configured	Must	✓ Done	vsftpd + dedicated user
Documentation covers installation & config steps	Must	✓ Done	This document
Password not in shell script (mysql_config_editor)	Should	✓ Done	~/.mylogin.cnf
MySQL user with minimum privileges for mysqldump	Should	✓ Done	PROCESS + SELECT + LOCK

Manual import procedure documented	Should	☑ Done	Section 7
Full OS backup at regular intervals	Should	⚠ Partial	Not yet scheduled
Backup files older than 7 days auto-deleted	Could	☑ Done	<i>find -mtime +7 in script</i>
FTP jail (chroot) configured	Could	☑ Done	<i>chroot_local_user=YES</i>
Backup logs (success/failure) written to log file	Could	☑ Done	<i>backup.log</i>

4. Virtual Machine & OS Setup

The creation and initial OS installation steps are not required to be documented per the work order. The following picks up after a clean Ubuntu Server 24.04 LTS installation.

4.1 Hostname & Network

Set the hostname to include the student IAM code:

```
sudo hostnamectl set-hostname sofjo685-backup
sudo nano /etc/hosts
# Add: 127.0.1.1    sofjo685-backup
```

Configure a static IP address by editing the Netplan file:

```
sudo nano /etc/netplan/00-installer-config.yaml

network:
  version: 2
  ethernets:
    ens18:
      dhcp4: no
      addresses: [10.12.2.X/24]    # IP assigned by teacher (didn't receive any)
      routes:
        - to: default
          via: 10.12.2.1
      nameservers:
        addresses: [10.12.2.1]

sudo netplan apply
```

4.2 System Updates & Timezone

```
sudo apt update && sudo apt upgrade -y
sudo timedatectl set-timezone Europe/Luxembourg
timedatectl    # verify
```

4.3 OpenSSH Server

```
sudo apt install -y openssh-server
sudo systemctl enable --now ssh
```

4.4 MySQL Client Software

```
sudo apt install -y mysql-client
```

5. Web & Database Server Configuration

Before the backup server can connect, the main server must be configured to allow remote MySQL connections and a dedicated backup user must be created.

5.1 Allow External MySQL Connections

Edit the MariaDB bind address so it listens on all interfaces (or the specific backup server IP):

```
sudo nano /etc/mysql/mariadb.conf.d/50-server.cnf

# Change:
bind-address = 0.0.0.0

sudo systemctl restart mariadb
```

Ensure UFW allows the MySQL port from the backup server:

```
sudo ufw allow from 10.12.2.XX to any port 3306
```

5.2 Create the Backup MySQL User

Log into MariaDB on the web server and create a user with only the privileges required by mysqldump:

```
sudo mysql -u root

CREATE USER 'backup_sofjo685'@'10.12.2.X'
  IDENTIFIED BY 'secure_password_here';

GRANT SELECT, PROCESS, LOCK TABLES, SHOW VIEW,
  EVENT, TRIGGER
  ON sofjo685_pif.*
  TO 'backup_sofjo685'@'10.12.2.X';

FLUSH PRIVILEGES;
EXIT;
```

Note: Replace 10.12.2.XX with the actual IP of the backup server, and choose a strong password.

6. Backup Directory & Permissions

```
sudo mkdir -p /srv/backup
sudo chown sofjo685:sofjo685 /srv/backup
sudo chmod 700 /srv/backup
```

Only the sofjo685 user has access. No other system users can read or write to this directory.

7. Backup Shell Script

7.1 Store Credentials Securely

Instead of hardcoding the password in the script, use `mysql_config_editor` to save credentials in an encrypted login file (`~/.mylogin.cnf`):

```
mysql_config_editor set \  
  --login-path=backup \  
  --host=10.12.2.X \  
  --user=backup_sofjo685 \  
  --password  
  
# Verify:  
mysql_config_editor print --login-path=backup
```

Replace 10.12.2.YY with the IP of the web & database server.

7.2 The Backup Script

Create the script at `/srv/backup/backup.sh`:

```
nano /srv/backup/backup.sh
```

```
#!/bin/bash  
# _____  
# backup.sh - Daily database backup for sofjo685_pif  
# Author : SOFJO685  
# Runs   : Daily at 08:00 via cron  
# _____  
  
BACKUP_DIR="/srv/backup"  
DB_NAME="sofjo685_pif"  
DATE=$(date +"%Y-%m-%d_%H-%M")  
FILENAME="${BACKUP_DIR}/${DB_NAME}_${DATE}.sql.gz"  
LOGFILE="${BACKUP_DIR}/backup.log"  
RETENTION_DAYS=7  
  
log() {  
    echo "[$(date '+%Y-%m-%d %H:%M:%S')] $1" >> "$LOGFILE"  
}  
  
log "INFO Starting backup of ${DB_NAME}"  
  
# Perform the dump - credentials read from ~/.mylogin.cnf  
if mysqldump --login-path=backup \  
    --single-transaction \  
    --routines --triggers \  
    "${DB_NAME}" | gzip > "$FILENAME"; then  
    log "INFO Backup succeeded: $FILENAME"  
else  
    log "ERROR Backup FAILED for ${DB_NAME}"  
    exit 1  
fi
```

```
# Remove backups older than RETENTION_DAYS days
DELETED=$(find "$BACKUP_DIR" -maxdepth 1 \
               -name "${DB_NAME}_*.sql.gz" \
               -mtime +${RETENTION_DAYS} -print -delete | wc -l)
log "INFO Deleted ${DELETED} old backup(s) (> ${RETENTION_DAYS} days)"

log "INFO Backup job finished"
```

Give the script execution permissions:

```
chmod +x /srv/backup/backup.sh
```

7.3 Schedule via cron

```
crontab -e

# Add the following line:
0 8 * * * /srv/backup/backup.sh
```

Verify the cron entry:

```
crontab -l
```

8. FTP Server Setup

8.1 Install vsftpd

```
sudo apt install -y vsftpd
```

8.2 Create FTP User

Create a dedicated user whose home directory is the backup directory:

```
sudo useradd -d /srv/backup -s /usr/sbin/nologin ftp_sofjo685
sudo passwd ftp_sofjo685
sudo chown sofjo685:ftp_sofjo685 /srv/backup
sudo chmod 750 /srv/backup
```

8.3 Configure vsftpd

```
sudo nano /etc/vsftpd.conf

# Relevant settings to set/verify:
listen=YES
anonymous_enable=NO
local_enable=YES
```

```
write_enable=NO          # read-only for FTP user
chroot_local_user=YES     # FTP jail - cannot go above /srv/backup
allow_writeable_chroot=NO
local_root=/srv/backup
pasv_enable=YES
pasv_min_port=40000
pasv_max_port=40100

sudo systemctl restart vsftpd
sudo systemctl enable vsftpd
```

The `chroot_local_user=YES` directive jails the FTP user inside `/srv/backup` — they cannot navigate to higher-level paths.

8.4 Open Firewall Ports

```
sudo ufw allow 21/tcp
sudo ufw allow 40000:40100/tcp
sudo ufw reload
```

9. Manual Database Restore Procedure

To restore a backup file to the MySQL database, follow these steps:

Step 1 — Copy the backup file to the database server

```
# From backup server to web server (replace paths/IPs as needed):
scp /srv/backup/sofjo685_pif_2026-04-17_08-00.sql.gz \
    sofjo685@10.12.2.YY:/tmp/
```

Step 2 — Decompress the file

```
gunzip /tmp/sofjo685_pif_2026-04-17_08-00.sql.gz
# Result: /tmp/sofjo685_pif_2026-04-17_08-00.sql
```

Step 3 — Import into MariaDB

```
sudo mysql -u root sofjo685_pif < /tmp/sofjo685_pif_2026-04-17_08-00.sql
```

If the database needs to be recreated first (e.g. after a crash):

```
sudo mysql -u root
DROP DATABASE IF EXISTS sofjo685_pif;
CREATE DATABASE sofjo685_pif CHARACTER SET utf8mb4 COLLATE utf8mb4_general_ci;
EXIT;

sudo mysql -u root sofjo685_pif < /tmp/sofjo685_pif_2026-04-17_08-00.sql
```

Step 4 — Verify the restore

```
sudo mysql -u root -e "USE sofjo685_pif; SHOW TABLES; SELECT COUNT(*) FROM measurement;"
```

10. Full OS Backup (Should)

For a complete backup of all operating system files, rsync can be used to copy the entire VM filesystem to an external location at regular intervals.

```
# Install rsync
sudo apt install -y rsync

# Example: backup to an NFS share or secondary storage
sudo rsync -aAXv --
exclude={'/dev/*','/proc/*','/sys/*','/tmp/*','/run/*','/mnt/*','/lost+found'}
\
  / /mnt/os-backup/

# Schedule weekly via cron (Sunday at 02:00):
# 0 2 * * 0 rsync -aAX ... >> /srv/backup/os_backup.log 2>&1
```

Note: Full OS backup scheduling is marked as **Should** and is partially implemented. The rsync command above is functional but requires a dedicated mount point to be configured first.

11. Gap Analysis — What Is Still Missing for the Presentation

The following items were identified by comparing this documentation against the REPIF1 documentation (sofjo685) and the L6 User Story requirements. These gaps should be addressed before the REPIF2 interim presentation.

11.1 Critical Gaps (Must — Not Yet Done)

- **No backup documentation in REPIF1 at all** — The REPIF1 document covers Raspberry Pi setup, the database, the web server and the website, but contains zero mention of the backup server. This entire section is new and must be added to or referenced from REPIF1.
- **REPIF1 does not document the remote MySQL user** — Section 5.2 of this document (creating backup_sofjo685 on the web server) has no equivalent in REPIF1. This configuration step on the main server must be documented.
- **REPIF1 does not document MariaDB bind-address change** — Allowing external connections on the web server (Section 5.1) is absent from REPIF1.

11.2 Should-Level Gaps

- **Full OS backup not yet scheduled** — The rsync command exists but has not been configured as a recurring cron job with a verified mount point. Must be completed before the presentation.
- **No verification / test run shown in documentation** — Neither the REPIF1 document nor this one shows evidence of running the backup script manually and confirming that a .sql.gz file was created and that the log entry was written.
- **No FTP connection test documented** — There is no screenshot or terminal output showing a successful FTP login as ftp_sofjo685 and directory listing.

11.3 Could-Level Gaps

- **Log rotation not configured** — The backup.log file will grow indefinitely. logrotate should be configured for /srv/backup/backup.log.
- **FTP jail testing not shown** — The documentation states chroot_local_user=YES is set but there is no proof (e.g. terminal output) demonstrating the user cannot cd above /srv/backup.

11.4 Summary Checklist for Presentation

Requirement	Priority	Status	Notes
Add backup server section to REPIF1 doc	Must	⚠ Partial	Only in this doc
Document MariaDB remote access on web server	Must	⚠ Partial	Not in REPIF1

Show manual script test run + log output	Should	 Partial	No evidence yet
Show FTP login test	Should	 Partial	No evidence yet
Schedule and verify full OS backup	Should	 Partial	Command ready only
Configure logrotate for backup.log	Could	 Partial	Not done